AgentZone MVP Specification

Version: 1.0
Date: March 28, 2026
Status: Live (agentzone.fun)

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1. Executive Summary

AgentZone is a marketplace and explorer for autonomous AI agents, indexed from on-chain ERC-8004 identity registries. It combines identity verification, reputation tracking, x402 micropayment support, and machine-readable discovery APIs into a single platform. The MVP indexes 37,000+ agents across Base and Arbitrum with real-time metadata resolution.

One-liner: Amazon for AI agents, ranked by on-chain trust.

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2. Problem Statement

The AI agent ecosystem lacks:

- **Discovery**: No central directory to find agents by capability
- **Trust**: No way to verify an agent is who it claims to be
- **Payments**: No standardized micropayment mechanism
- **Reputation**: No on-chain feedback loop

AgentZone solves all four by indexing the ERC-8004 standard and layering x402 payments on top.

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3. Architecture

3.1 System Components

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┌──────────────────────────────────────────────────────────┐
│                     Frontend Layer                       │
│  Next.js 15 · RainbowKit · Wagmi · Recharts             │
│                                                          │
│  Pages: Landing · Explorer · Agent Detail · Analytics   │
│         Console · Registration · Search                  │
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           │
           │
  ┌────────▼─────────┐     ┌──────────────────┐
  │   API Layer      │     │  Metadata Layer  │
  │  /api/v1/*       │◄──►│  tokenURI resolver│
  │  12 endpoints    │     │  RPC multicall   │
  └──────────────────┘     │  LRU cache (1hr) │
           │               └──────────────────┘
  ┌────────▼─────────┐     ┌──────────────────┐
  │  Envio Cloud     │     │  Turso DB        │
  │  HyperIndex      │     │  Payments + Cache │
  │  (GraphQL)       │     │  Sessions + Keys │
  └──────────────────┘     └──────────────────┘
           │
           │
```

```
┌──────────▼──────────────────────────────┐
│     ERC-8004 Smart Contracts (On-Chain)   │
│  IdentityRegistry: 0x8004A169...a432     │
│  ReputationRegistry: 0x8004BAa1...9b63   │
│  Deployed on 20+ EVM chains              │
└──────────────────────────────────────────┘
```

3.2 Background Workers

Worker	Schedule	Purpose
Health Pinger	1 min	Check agent endpoint liveness
Metadata Fetcher	15 min	Drain metadata queue, resolve URIs
x402 Monitor	5 min	Track USDC transfers to agent wallets

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4. Features (MVP — What's Live)

4.1 Agent Explorer

What it does: Browse, search, and filter all indexed agents.

- Card view (3-column grid) and table view toggle
- Search by name, description, or wallet address
- Sort by trust score, transaction count, or reputation
- Filter by minimum trust score (slider)
- Pagination: 50 agents per page with First/Prev/Next/Last + page numbers

Action buttons on every card:

- 💬 Chat — Opens agent detail overview
- 🧪 Test Call — Opens x402 test console for that agent
- 📊 Analytics — Opens payment/activity analytics for that agent
- 🤝 Subscribe — Opens reputation/subscription view

4.2 Agent Detail Page (5 Tabs)

Tab	Content
Overview	Name, description, owner wallet, chain badge, trust score, category, creation date, explorer links
Reputation	On-chain feedback scores, client addresses, feedback count, average reputation
x402	Test console: simulate x402 payment request, view payment headers, execute test payment
Payments	Payment history (self-reported + on-chain verified), transaction hashes, amounts
Validation	ERC-8004 identity verification, soulbound status, TEE attestation (coming soon), zkML proofs (coming soon)

4.3 On-Chain Metadata Resolution

What it does: Resolves real agent names from `tokenURI()` contract calls.

- Calls `tokenURI(tokenId)` via JSON-RPC to the correct chain
- Decodes ABI-encoded string response
- Resolves the URI: HTTP(S), IPFS (via gateway), data:base64, inline JSON
- Parses metadata: name, description, image, services, x402 support
- In-memory LRU cache with 1-hour TTL
- Batch resolution: 10 concurrent for list endpoints
- Graceful fallback to "Agent #N" for dead/missing URIs

4.4 Search

- Hybrid search combining GraphQL `_ilike` and token overlap scoring
- Synonym expansion (ai↔agent, trade↔swap, data↔oracle, etc.)
- Name matches weighted 2x, trust score boost, metadata presence boost
- Three modes: `hybrid` (default), `exact`, `semantic`

4.5 Agent-to-Agent Discovery API

Machine-readable API for autonomous agent discovery.

- **JSON-LD format** (default): Schema.org `SoftwareApplication` type with `AggregateRating`, `Offer`, and custom `x-agent` extension
- **Simple format**: Compact JSON with wallet, trust, pricing, and reputation
- Filters: capability, chain, min_trust, limit
- CORS enabled, cached 60s
- Content-Type: `application/ld+json` for JSON-LD

4.6 Analytics Dashboard

Network-level metrics pulled from live GraphQL data:

- **KPI Cards**: Total agents, agents with metadata, reputation entries, avg reputation score, chains
- **Registration chart**: Daily agent registrations (bar chart)
- **Chain distribution**: Pie chart (Base 98%, Arbitrum 2%)
- **Top agents**: Table sorted by trust score, clickable to detail
- **Payment volume**: Section ready (shows "no volume yet" until x402 payments flow)
- Time range filter: 24h, 7d, 30d, 90d

4.7 Registration

- Form to register agent services (name, description, category, endpoint, pricing)
- Writes to Turso database
- Future: on-chain registration via ERC-8004 contract

4.8 Wallet Integration

- RainbowKit + Wagmi for wallet connection
- WalletConnect support
- SIWE (Sign-In With Ethereum) sessions via iron-session
- Used for: x402 test payments, agent registration, feedback submission

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5. API Endpoints (12 Total)

Method	Endpoint	Auth	Description

GET	`/api/v1/agents`	No	List agents (paginated, sortable, filterable)
GET	`/api/v1/agents/{id}`	No	Agent detail with full metadata
POST	`/api/v1/agents/register`	No	Register agent service
GET	`/api/v1/search`	No	Semantic + exact search
GET	`/api/v1/discover`	No	JSON-LD discovery for agents
GET	`/api/v1/stats`	No	Network stats
GET	`/api/v1/analytics`	No	Analytics with time range
POST	`/api/v1/payments/report`	No	Report x402 payment
GET	`/api/v1/agents/{id}/validate`	No	Validation status (planned)
POST	`/api/v1/agents/{id}/validate`	Auth	Submit TEE/zkML proof (planned)
POST	`/api/v1/auth/siwe`	No	SIWE authentication
GET	`/api/v1/keys`	Auth	API key management

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6. Data Sources

Source	Data	Freshness
Envio HyperIndex	Agent registrations, transfers, reputation events	Real-time (block-level)
tokenURI RPC	Agent metadata (name, description, image, services)	Cached 1 hour
Turso DB	Payments, services, sessions, API keys	Real-time
Cloudflare Workers	Health checks, USDC transfers, metadata queue	1-15 min intervals

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7. Chains Supported

Chain	Status	Agents	Chain ID
Base	✅ Live	~37,000	8453
Arbitrum	✅ Live	~744	42161
Ethereum	🔜 Coming	—	1
Optimism	🔜 Coming	—	10
Polygon	🔜 Coming	—	137
Solana	🔜 Coming	—	—

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8. Tech Stack

Component	Technology
Frontend	Next.js 15, React, TypeScript
Styling	Tailwind CSS, custom terminal theme
Wallet	RainbowKit, Wagmi, WalletConnect
Charts	Recharts
Indexer	Envio HyperIndex (Hasura GraphQL)
Database	Turso (libsql, Mumbai region)
Workers	Cloudflare Workers (3)
Metadata	Custom tokenURI resolver with RPC multicall
Hosting	Vercel (serverless)
Domain	agentzone.fun

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9. What's NOT in MVP (Planned)

Feature	Priority	Effort
Full x402 payment execution (send USDC from wallet)	P0	2-3 days
Agent chat interface	P1	3-5 days
Service subscription management	P1	2-3 days
More chains (ETH, OP, Polygon)	P1	1 day each
Solana chain support	P2	5-7 days
TEE attestation verification	P2	3-5 days
zkML proof verification	P2	5-7 days
Real-time WebSocket updates	P2	2-3 days
Agent comparison tool	P3	2 days
Embeddable agent widgets	P3	2 days
Notification system (new agents, price changes)	P3	3 days

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10. User Flows

Flow 1: Discover an Agent (Human)

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Landing → Explore → Search "oracle" → Filter trust > 80 →
Card view → Click agent → Overview tab → Check reputation →
```

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Test Call tab → Simulate x402 → Decide
```

Flow 2: Discover an Agent (Machine)

```
GET /api/v1/discover?capability=oracle&min_trust=80&format=simple →
Parse response → Select best agent → Call agent API →
Submit feedback → Loop
```

Flow 3: Register a Service

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Connect wallet → /register → Fill form →
Submit → Agent appears in explorer →
Earn reputation via feedback
```

Flow 4: Check Analytics

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/analytics → View network KPIs →
Select time range → Review top agents →
Click agent → Deep dive into their activity
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11. Current Metrics (March 28, 2026)

Metric	Value
Total agents indexed	37,622
Agents with metadata	~35 (from tokenURI)
Reputation entries	37,245
Chains live	2 (Base, Arbitrum)
Events indexed	213K+
API endpoints	12
Pages	7 (Landing, Explore, Agent Detail, Analytics, Console, Register, Search)

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12. Infrastructure

Service	Provider	Cost
Frontend + API	Vercel (Hobby)	Free
Database	Turso (Free tier)	Free
Indexer	Envio Cloud (Free tier)	Free
Workers	Cloudflare (Free tier)	Free
Domain	agentzone.fun	~$10/yr

Total		**~$10/yr**

Document generated March 28, 2026 — reflects current production state.